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May 31, 2026

Hiring Committee

Google Student Researcher Program

Google LLC

Dear Google Student Researcher Hiring Committee,

I am writing to apply for the **Google Student Researcher (BS/MS) Program**. I am a final-year Computer Engineering student at Bahauddin Zakariya University, Multan, Pakistan (CGPA: 3.39/4.0, graduating May 2026), with hands-on experience building AI-integrated systems and a strong foundation in machine learning, mobile application development, and research. The opportunity to contribute to open-ended, exploratory research alongside Google's scientists is exactly the kind of challenge I have been working toward.

My final year project, **Zenvyra Health**, is a complete AI health monitoring system that combines an ESP32 wearable with six biomedical sensors (heart rate, SpO2, temperature, GSR, ECG, and motion) with a Kotlin/Jetpack Compose Android application. I trained and deployed a **TensorFlow Lite on-device stress classification model** that processes real-time multi-sensor data streams, designed the full Firebase backend for secure data management and PDF health report generation, and built wellness features including AI-driven guided breathing, streak tracking, and an in-app coaching chatbot. This project required me to independently bridge hardware, ML, and software engineering — which I believe aligns directly with the kind of multi-disciplinary, research-oriented work Google's teams pursue.

Alongside this, I am building **HireFlow**, an AI recruitment platform that implements a multi-layer NLP pipeline: TF-IDF keyword extraction, KeyBERT skill identification, cosine similarity-based semantic resume-to-job-description matching, and voice sentiment analysis for interview response evaluation. I am integrating the **Gemini API** for intelligent JD parsing and AI-generated candidate feedback. These projects reflect my genuine interest in applied ML research and in building systems where AI creates measurable human impact.

Beyond software, I bring over two and a half years of structured research experience — producing 100+ evidence-based papers and analytical reports across technology, healthcare, and social sciences —

combined with quantitative analysis expertise in SPSS and qualitative analysis in NVivo. I have also contributed to AI output evaluation work on platforms like Prolific and Scale AI, which has sharpened my ability to think critically about model behavior, bias, and human-aligned system design. I am proficient in Python, Kotlin, JavaScript, C++, and have experience with deep learning frameworks and semantic NLP techniques.

Google's research culture — open-ended problems, flexibility in approach, collaboration with leading researchers — is the environment where I believe I can contribute most meaningfully. I am particularly drawn to research areas intersecting **machine learning, human-computer interaction, and AI for social good**, themes that run through both my technical projects. I am eager to bring my engineering background, research mindset, and genuine curiosity to a Google research team and to grow through the mentorship this program offers.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my background aligns with your current research needs.

Sincerely,

Noor Fatima

Bachelor of Computer Engineering — Bahauddin Zakariya University, Multan

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